

MCQ related to heme, liver, vitamins, minerals, nutrition and other.
Please take the 50 MCQ in 75 min and read the explanations!
The questions and explanations are provided by SGU SOM Biochemistry faculty.

Q#1	During your clinical rotations, one of your patients is a 25-year-old woman. Her eyes show “Kayser-Fleischer rings”. Which defect related to mineral metabolism leads to her condition? - Copper deficiency - Toxic copper levels - Iron deficiency - Toxic iron levels - Zinc deficiency
Q#2	Anemia can be grouped into normocytic, microcytic and macrocytic anemia. The deficiency of which of the following vitamins leads to microcytic anemia? - Thiamine (vitamin B1) - Riboflavin (vitamin B2) - Niacin (vitamin B3) - Pyridoxine (vitamin B6) - Cobalamin (vitamin B12)
Q#3	The blood clotting cascade works via proteolytic activation of blood clotting factors. It is critical that the cascade is performed in a fast and efficient manner. For this reason, specific inactive blood clotting factors are γ-carboxylated in the liver (maturation of blood clotting factors), so that they can form calcium bridges to activated platelets. What is special about this process? The process of γ-carboxylation - uses as coenzyme phyloquinone and/or menaquinone - is activated by the drug warfarin in former stroke patients - is activated during severe liver damage in chronic alcoholics - uses biotin as coenzyme for the incorporation of carbon dioxide - can totally be prevented by a the diet low in phyloquinone
Q#4	An abnormality of copper metabolism can lead to severe health problems. Which of the following defects and diseases are related to low levels of ceruloplasmin and high toxic copper levels? - Aneurisms - Menke’s syndrome - Rickets - Blue sclerae - Wilson’s disease.
Q#5	The lipid soluble vitamins A and D lead to molecules that have steroid-hormone like actions in the nucleus. Vitamin A is part of the group of retinoids. Which retinoid is formed inside of cells in the cytosol and has in the nucleus steroid hormone-like action related to transcription? - Retinol - Beta-carotene - Retinoic acid - Retinal - Cis-retinal

Q#6	A two year-old child is brought by his parents to the family physician. The child has soft, pliable bones and has difficulty to stand. The parents explain that the child did not like the taste of milk and cheese, and that they had kept him inside and away from sunlight as his skin was easily burnt by the sun. The physician diagnoses rickets and prescribes vitamin treatment with vitamin D. The physiological active form of vitamin D is calcitriol which has hormone-like action related to transcription. Where and how is calcitriol formed in the human body? In the a. skin from 7-dehydrocholesterol b. liver from dietary vitamin D c. blood from 25-OH D d. kidney from 25-OH D e. intestinal cells from cholesterol
Q#7	The moderate deficiency of vitamin B6 leads to general tiredness, irritability and depression. Vitamin B6 is used to form the coenzyme pyridoxal phosphate which is needed by many enzymes. Which of the following enzymes needs pyridoxal phosphate as coenzyme? a. ALA dehydratase b. Methionine synthase c. Transketolase d. Aminotransferase e. Methylmalonyl-CoA mutase
Q#8	Ascorbic acid is a vitamin (Vitamin C) in humans, as it cannot be formed in primates. A severe deficiency can lead to scurvy which is characterized by sore gums, loose teeth or loss of teeth, and fragile blood vessels. What is a major function of Vitamin C in humans? Vitamin C is needed for a. dietary heme iron uptake b. allysine formation for elastin and collagen cross-linking c. hydroxyprolyl formation during collagen synthesis d. prevention of hemorrhagic disease of the newborn e. formation of tyrosine from phenylalanine
Q#9	Selenium is a trace mineral that is not needed as cofactor for many enzymes. Which of the following enzymes needs selenium as cofactor? a. Superoxide dismutase b. Catalase c. Glutathione reductase d. Glutathione peroxidase e. Nitric oxide synthase
Q#10	The enzyme xanthine oxidase catalyzes the reactions from hypoxanthine to xanthine and also from xanthine to uric acid. Which of the following trace minerals is needed as cofactor in both reactions? a. Chromium b. Molybdenum c. Manganese d. Zinc e. Copper

Q#11	Biotin is a vitamin grouped in the B-complex as energy-releasing vitamin. Biotin is covalently bound to a lysine residue of specific enzymes, and then it is often referred to as biocytin. Which of the following enzymes needs biotin as prosthetic group? - Acetyl-CoA carboxylase - Carbamoyl-phosphate synthase I - Enzymes performing γ-carboxylations - Methylmalonyl-CoA mutase - Pyruvate dehydrogenase
Q#12	Methionine contains in its structure a methyl-group. In the activated form of methionine as S-adenosyl-methionine (SAM) this methyl group is used by many enzymes for methylation of their substrates. Which of the following compounds is methylated using SAM? - Homocysteine - Dopamine - Norepinephrine - dUMP - Tetrahydrofolate
Q#13	The trace mineral manganese is not needed as cofactor for many enzymes. Which of the following enzymes needs manganese as cofactor? - Pyruvate carboxylase - Cytosolic superoxide dismutase - Dopamine hydroxylase - Tyrosinase - Carbonic anhydrase
Q#14	A 38-year old male changes his occupation and the new employer insists on a health check-up. The physician notices an abnormal slate-grey color of his skin. The blood analysis shows elevated levels of ALT, AST and ALP. The blood glucose level is 200 mg/dL. The physician suspects hereditary hemochromatosis and she orders additional blood tests. Which abnormalities at the biochemical level are found in patients with hemochromatosis? - High serum ferritin levels - Deficiency of ceruloplasmin (ferroxidase) - Depleted cellular hemosiderin levels - Reduced dietary iron uptake - Decreased transferrin in the blood
Q#15	Vitamin B12 deficiency can be due to a reduced synthesis of intrinsic factor (IF) in the elderly. What is special about the intrinsic factor? - It is secreted by chief cells in the stomach - The IF-B12 complex is degraded by pancreatic proteases

	c. It transports vitamin B12 in the blood d. It is synthesized in intestinal mucosal cells e. It is needed for uptake of B12 into intestinal mucosal cells
Q#16	A 40 year-old woman was admitted to the emergency room. She was in a highly agitated state, weeping and complaining about severe abdominal pain. After talking to the patient, the physician thought that his patient could suffer from Acute Intermittent Porphyria and he ordered a rapid urine test for porphobilinogen. After detection of porphobilinogen the urine was further analyzed and showed also increased levels of ALA. What would be the most appropriate treatment? Administration of a. a blood transfusion b. vitamin B₆ c. barbiturates d. hemin e. drugs against photosensitivity
Q#17	Which of the following injury markers in the blood is the most specific for liver damage due to ethanol abuse? Increased levels of a. Alanine aminotransferase (ALT) b. Aspartate aminotransferase (AST) c. Alkaline phosphatase (ALP) d. Albumin e. γ-glutamyl transpeptidase (GGT)
Q#18	Heme degradation involves many different cells like macrophages, hepatocytes, intestinal bacteria and kidney cells. A defect in this catabolism can lead to jaundice, which is due to high bilirubin levels in the blood (>3mg/dL). What happens normally to bilirubin that is released from the reticuloendothelial system? Bilirubin a. is transported in blood bound to ligandin b. is taken into the liver and conjugated to bilirubin-diglucuronide c. can lead to kernicterus in an adult with jaundice d. is degraded in the fetus to urobilinogen e. is converted to urobilinogen in the hepatocyte
Q#19	Heme synthesis takes place mainly in the erythroid cells and in the hepatocytes. The synthesis starts and ends in mitochondria with cytosolic enzymes in between. What is the regulated enzyme of heme synthesis in the hepatocyte? a. ALA dehydratase b. ALA synthase c. Uroporphyrinogen decarboxylase d. Ferrochelatase e. Uroporphyrinogen III synthase
Q#20	Jaundice can result from genetic defects or from acquired liver or biliary disease. One can group jaundice into prehepatic, hepatic, or posthepatic jaundice. What is characteristic for prehepatic jaundice?

	a. Gallstones block the biliary tracts b. The levels of alkaline phosphatase are elevated in blood c. The urine is of a reddish brown color due to bilirubin in urine d. It leads to high levels of conjugated bilirubin in blood e. It can occur following a massive lysis of RBC
Q#21	Ethanol metabolism affects the whole body. Most of the ethanol reaches the liver and is metabolized in hepatocytes to acetaldehyde and acetate. Which of the following changes of liver metabolism occurs at a high ethanol concentration in the liver? a. Increased synthesis of triacylglycerols b. Inhibition of ketone body formation during fasting c. Decreased NADH/NAD⁺ ratio d. Inhibition of glycogen degradation e. Increased uptake of lactate for gluconeogenesis
Q#22	The liver performs many pathways after a meal and during fasting. Which of the following pathways is not performed in the liver after two days of fasting? a. Gluconeogenesis b. Glycogen degradation c. Ketone body formation d. Urea cycle e. Fatty acid degradation
Q#23	The effects of ethanol on liver metabolism depend on the actual ethanol concentration in the hepatocyte and on the fact whether chronic ethanol abuse is present. What happens in the liver after a high ethanol intake by a person who is not a chronic-alcoholic? a. Development of temporary hyperlipidemia b. Development of a fatty liver c. Development of liver cirrhosis d. Activation of gluconeogenesis due to low cytosolic NADH levels e. Inhibition of ketone body formation
Q#24	Porphyrias are diseases related to heme synthesis and their clinical symptoms often depend on the accumulation of intermediates following the deficiency of specific enzymes. What is characteristic for porphyria cutanea tarda? This porphyria a. leads to accumulation of protoporphyrin IX b. leads to patients who are highly irritable but who are not photosensitive c. is the result of the deficiency of uroporphyrinogen decarboxylase d. is the result of lead poisoning e. is a very rare porphyria
Q#25	Jaundice is the result of elevated bilirubin levels in the blood. A mild jaundice in newborns or premature babies is common and is referred to as

	a. Crigler-Najjar syndrome I b. Obstructive jaundice c. Gilbert's syndrome d. Physiological jaundice e. Arias syndrome
Q#26	The liver performs many pathways that are also found in other cells. But the liver performs also pathways that are unique for hepatocytes. Which of the following synthetic pathways is unique for the liver? a. Gluconeogenesis b. Glycogen synthesis c. Fatty acid synthesis d. Amino acid synthesis e. Ketone body synthesis
Q#27	You saw a picture of a patient with jaundice, and it was very clear how the yellow color showed up in the sclerae of his eyes due to accumulation of a specific intermediate of total heme degradation. Which of the following compounds is found in high levels in the blood and accumulates in skin and sclerae in a patient with hepatocellular jaundice? a. Urobilinogen b. Bilirubin c. Urobilin d. Stercobilin e. Biliverdin
Q#28	Heme degradation is a process that involves several tissues and also bacteria. What is characteristic for heme degradation? It leads to a. formation of carbon monoxide by heme oxygenase b. conjugation of biliverdin with UDP-glucuronate in the liver c. bilirubin that is covalently bound to albumin in blood d. conjugated bilirubin that is formed in the gallbladder e. brown urobilinogen in the large intestine
Q#29	During fasting, the liver uses β-oxidation of fatty acids to generate eventually ATP and acetyl-CoA, which are both needed for gluconeogenesis. Carbons of fatty acids cannot be used for gluconeogenesis, but acetyl-CoA formed in β-oxidation is needed as a. absolute activator for pyruvate carboxylase b. absolute activator for PEP carboxykinase c. substrate for acetyl-CoA carboxylase d. substrate for cytosolic HMG-CoA synthase e. inhibitor of the carnitine shuttle

Q#30	Enzyme induction is a tool to increase the actual amount of the enzyme. The hormones insulin and glucagon favor different pathways in the liver and induce different key enzymes. Which of the following enzymes is induced in the liver by insulin? a. Glucose-6-phosphatase b. Hexokinase c. Glucokinase d. PEP carboxykinase e. Fructose-1,6-bisphosphatase																					
Q#31	A 22-year old female student was brought by her friends to the emergency room on New Years Day. She was sweating profusely, hyperventilating and scared by irregular heart beats. She was highly agitated and her breath smelled of alcohol. Her friends told the physician, that she had been drinking heavily during the whole night. She did not take medications and had slept for several hours. She suddenly woke up from unusual fast heartbeats. Her lab results showed the following <table><tr><td>Aspartate aminotransferase (AST)</td><td>30 U/L</td><td>(reference range 5-35)</td></tr><tr><td>Alanine aminotransferase (ALT)</td><td>50 U/L</td><td>(reference range 5-56)</td></tr><tr><td>Alkaline phosphatase (ALP)</td><td>70 U/L</td><td>(reference range 20-160)</td></tr><tr><td>γ-glutamyl transpeptidase (GGT)</td><td>157 U/L</td><td>(reference range 8-78)</td></tr><tr><td>Total bilirubin</td><td>1.4 mg/dL</td><td>(normal <1.4 mg/dL)</td></tr><tr><td>Direct bilirubin</td><td>normal</td><td></td></tr><tr><td>Albumin</td><td>4.3 g/dL</td><td>(reference range 3.5-5.5)</td></tr></table> Which of the following is most likely in the clinical scenario? a. Liver cirrhosis b. Lactic acidemia c. Hypoketonemia during fasting d. Obstructive jaundice e. Hepatocellular damage	Aspartate aminotransferase (AST)	30 U/L	(reference range 5-35)	Alanine aminotransferase (ALT)	50 U/L	(reference range 5-56)	Alkaline phosphatase (ALP)	70 U/L	(reference range 20-160)	γ-glutamyl transpeptidase (GGT)	157 U/L	(reference range 8-78)	Total bilirubin	1.4 mg/dL	(normal <1.4 mg/dL)	Direct bilirubin	normal		Albumin	4.3 g/dL	(reference range 3.5-5.5)
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Q#32	A 45 year old man complains about steatorrhea. The sclerae of his eyes are yellowish. Blood analysis shows high levels of direct bilirubin and increased alkaline phosphatase (ALP). The blood levels of alanine aminotransferase (ALT), aspartate aminotransferase (AST), albumin and inactive blood clotting factors are normal. The dipstick method shows bilirubin in the urine. What is most likely the reason for his jaundice? a. Gallstones b. Excessive hemolysis																					

	c. Gilbert's syndrome d. Crigler-Najjar Type II e. Severe liver damage
Q#33	Ethanol metabolism in the liver starts with alcohol dehydrogenase. This enzyme has a large variety of activity dependent on genetic background. What is a general characteristic for alcohol dehydrogenase? This enzyme a. forms NADH in mitochondria b. has a larger K_m for ethanol than CYP 2E1 c. forms acetaldehyde in cytosol d. forms acetate in mitochondria e. is part of MEOS, the microsomal ethanol oxidizing system
Q#34	Hemoglobin in mature RBC was synthesized in erythroid cells. This included heme synthesis while mitochondria were still available. What is special of heme synthesis in erythroid cells? It a. starts and ends in the cytosol b. is inhibited by heme c. leads to synthesis of globin chains d. is not inhibited by lead e. leads to hemin catalyzed by ferrochelatase
Q#35	MCAD deficiency is the most common genetic defect in fatty acid metabolism. The acute symptoms are similar to Reye syndrome and SIDS. What is the most specific observation found in patients with MCAD deficiency which is not found in other diseases? a. Ketoacidosis b. Hypoglycemia c. Dicarboxylic acids in urine d. Inhibition of CPT II e. Hyperglycemia and hypoketonemia
Q#36	A 40-year old woman is admitted to the emergency room. She fell down the stairs and still smells of alcohol. The neighbor who alerted the emergency team pointed out that the patient severely abused alcohol for several years. The attending physician decides to administer vitamins as part of her treatment of chronic alcoholism. The cofactor pyridoxal phosphate is formed from vitamin B₆ and is needed as coenzyme for a. GABA formation from glutamate b. methionine formation from homocysteine c. deoxyribonucleotide formation from ribonucleotides d. purine nucleotide degradation e. degradation of catecholamines
Q#37	During your clinical rotations, you are asked to take the medical history of a 3-year old boy. You come to the conclusion that this patient might suffer from Tay-Sachs disease. The reasoning for this genetic defect is the origin of his parents, specific onion-shell-like accumulations in his cells and that he shows

	a. hypopigmentation b. a cherry-red macula c. a blue sclera d. signs of self-mutilation e. a dislocated lens
Q#38	Which of the following compounds is needed for dopa decarboxylase ? a. copper and vitamin C b. tetrahydrofolate c. pyridoxal phosphate d. thiamine pyrophosphate e. SAM
Q#39	A 2 year-old girl is treated with enzyme replacement therapy by intravenous infusion in the hospital. She is short of stature, has coarse facial features and shows macrocephaly and hepatosplenomegaly. Her joint mobility is severely restricted. The modified enzyme that is used in her enzyme replacement therapy is iduronidase, which is given to patients with a. Gaucher Disease b. Hunter Syndrome c. Tay-Sachs Disease d. Niemann-Pick Disease e. Hurler Syndrome
Q#40	A 14 year old boy plans to travel with his parents to Egypt and he comes to the clinic for a routine check-up. On this occasion, he tells the physician that he has trouble sweating. On physical examination, the boy is found to have symptoms that are consistent with Fabry's disease, especially as he has dark reddish skin lesions with bathing trunk distribution. What would most likely pathologically accumulate in his vascular endothelium? a. galactocerebrosides b. globosides c. glucocerebrosides d. sphingomyelin e. ganglioside GM₂
Q#41	A 15 year old girl has been admitted to the Burns unit. She has had severe burns when there was an accidental fire in the house. She has been on fluid and electrolytes on admission and her blood pressure has been stabilized. On the third day in the hospital, which of the following metabolic changes would you expect? A. A decrease in circulating cortisol B. An increase in ketone body production C. Inhibition of hormone sensitive lipase in adipose tissue D. An increased rate of muscle proteolysis E. A decrease in the metabolic rate
Q#42	An obese subject has been diagnosed to have the metabolic syndrome (syndrome X). Which of the following features would be present in the subject?

	A. Pear shaped distribution of body fat B. Low levels of circulating insulin C. Increased blood triglyceride levels D. BMI of 24.6 E. Increased activity of lipoprotein lipase
Q#43	A child is suspected as having kwashiorkor. Which of the following parameters would be diagnostic of kwashiorkor? A. BMI greater than 25 B. Low serum albumin levels C. Increased waist circumference D. Increased blood urea levels E. Increased skin fold thickness
Q#44	A 20 year old female model is practically starving for the past 15 days to appear for a beauty pageant. She has lost 10 pounds over the past 10 days. She has lost consciousness and has been brought to the ER. Regarding the metabolic processes at this time (compared to the postabsorptive phase), the TRUE statement is A. Adipose tissue hormone sensitive lipase would be inhibited B. The insulin glucagon ratio would be increased C. The major source of gluconeogenesis would be lactate D. The metabolic rate would be increased E. The production of 3-hydroxybutyrate would be increased
Q#45	A 60 year old man on routine check-up was found to have an elevated serum cholesterol level. Which would be the most appropriate nutritional advice? A. Increase the saturated fatty acid content in the diet B. Decrease the content of arachidonic acid in the diet C. Increase the intake of ω-3 fatty acids in the diet D. Decrease the content of oleic acid in the diet E. Decrease the dietary fiber content
Q#46	A 17 year old boy has been admitted for the evaluation of weight loss over the past few weeks and the complaints of increased frequency of urination at night. Fasting blood glucose levels were elevated on two consecutive occasions. A presumptive diagnosis of type I diabetes mellitus has been considered. Which of the following metabolic features would support the diagnosis? A. Presence of insulin resistance B. Presence of high circulating proinsulin levels C. Low levels of circulating insulin D. Increased circulating leptin levels E. Low levels of glycated hemoglobin
Q#47	A 40 year old man has been brought to the ER in the unconscious state. He has a daughter diagnosed to have type I diabetes mellitus and on maintenance insulin therapy. The physician suspects that the man

	has intentionally induced hypoglycemia by insulin injections. Which of the following lab results would indicate that the hypoglycemia is factitious? A. Increased circulating insulin levels B. Increased proinsulin levels C. Decreased C peptide levels D. Presence of sulfonylurea in circulation E. Increased ketone bodies in urine
Q#48	A 30-year-old woman is in the hospital and her serum analysis shows elevated levels of α-fetoprotein. This protein a. Is part of the α_2-globulin fraction b. Is synthesized by activated B-cells c. Is a serum marker for liver cancer d. Is increased in patients with multiple myeloma e. Is one of the largest serum protein
Q#49	A war prisoner is on a hunger strike and reaches 40 days of starvation. Which of the following fuel sources is mostly used in his the brain at this time? a. Free fatty acids b. 3-hydroxybutyrate c. Acetoacetate d. Glucose e. Amino acids
Q#50	Which of the following serum levels could best indicate chronic liver disease/ cirrhosis? a. Decreased albumin b. Increased ALT c. Decreased ALP d. Decreased γ-globulins e. Increased α_2-macroglobulin